Complex Numbers

1. Simplify using D' Moivres Theorem.

a. $\frac{(\cos 5x - i\sin 5x)^2(\cos 7x + i\sin 7x)^{-3}}{(\cos 4x - i\sin 4x)^9(\cos x + i\sin x)^5}$ b. $\frac{(1+i)^6(\sqrt{3}-i)^4}{(1-i)^6(1+i\sqrt{3})^9}$ c. $\left(\frac{1+\sin x + i\cos x}{1+\sin x - i\cos x}\right)^n$

2. Using D'Moivres theorem prove that $\left(1+i\sqrt{3}\right)^8 + \left(1-i\sqrt{3}\right)^8 = -2^8$.

3. Using D'Moivres theorem prove that $\left(\sqrt{3}+i\right)^n + \left(\sqrt{3}-i\right)^n = 2^{n+1}\cos\frac{n\pi}{6}$.

4. Prove using D'Moivres theorem

a. $\frac{\sin 7x}{\sin x} = 7 - 56sin^2x + 112sin^4x - 64sin^6x$

b. $\frac{\sin 6x}{\sin 2x} = 16cos^4x - 16cos^2x + 3$

5. If $\sin 6x = a\cos^5 x \sin x + bcos^3x \sin^3 x + c\cos x \sin^5 x$, find the values of a, b, and c.

6. Prove using D' Moivres Theorem

a. $cos^7x = (\cos 7x + 7\cos 5x + 21\cos 3x + 35\cos x)/2^6$

b. $cos^5xsin^3x = -(\sin 8x + 2\sin 6x - 2\sin 4x - 6\sin 2s)/2^7$

7. Find all the values of the following using D' Moivres Theorem

a. $\left(\frac{1}{2} - i\frac{\sqrt{3}}{2}\right)^{\frac{1}{4}}$ b. $\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)^{\frac{3}{4}}$

8. Solve $x^4 - x^2 + 1 = 0$ using D'Moivres Theorem.

9. Show that the continued product of all the values of $i^{2/3}$ is –1.

10. If $\alpha, \alpha^2, \alpha^3, \alpha^4$ are the roots of $x^5 - 1 = 0$ then find α and prove that $(1-\alpha)(1-\alpha^2)(1-\alpha^3)(1-\alpha^4) = 5$.

11. Show that the sum of the n, nth roots of unity is zero and product is $(-1)^{n-1}$.

12. Solve the equation $x^9 + 8x^6 + x^3 + 8 = 0$.

13. Solve the equation $z^4 = i(z-1)^4$ and show that the real part of all the roots is ½.

14. If α, β are the roots of $z^2 \sin^2\theta - z.\sin 2\theta + 1 = 0$, then prove that $\alpha^n + \beta^n = 2\cos n\theta \, cosec^n\theta$.

15. If $\cos\alpha + \cos\beta + \cos\gamma = 0$, and $\sin\alpha + \sin\beta + \sin\gamma = 0$ prove that
$(i)\ \cos 3\alpha + \cos 3\beta + \cos 3\gamma = 3\cos(\alpha+\beta+\gamma)$
$(ii)\ \sin 3\alpha + \sin 3\beta + \sin 3\gamma = 3\sin(\alpha+\beta+\gamma)$

Hyperbolic Functions

12. Relationship between Circular and Hyperbolic functions.

a. $\sinh ix = i \sin x$ b. $\sin ix = i \sin hx$

c. $\cosh ix = \cos x$ d. $\cos ix = \cosh x$

13. Identities in Hyperbolic functions.

a. $e^x = \cosh x + \sinh x$ b. $\cosh^2 x - \sinh^2 x = 1.$

c. $\sinh(-x) = -\sinh x.$ d. $\cosh(-x) = \cosh x.$

14. Solve the following equation for real values of x.

a. $17 \cosh x + 18 \sinh x = 1.$

b. $6 \sinh x + 2\cosh x + 7 = 0$

15. Prove the following.

a. $\frac{1}{1-\frac{1}{1-\frac{1}{1+sinh^2 x}}} = -\sinh^2 x$

b. $\left(\frac{\cosh x+\sin hx}{\cos hx-\sinh x}\right)^n = \cosh 2nx + \sinh 2nx.$

c. $(\cosh x + \sinh x)^n = \cosh nx + \sinh nx.$

d. If $\cos\alpha \cosh\beta = (x/2)$ and $\sin\alpha.\sinh\beta = (y/2)$ then show that $\text{Sec}(\alpha - i\beta) + \text{Sec}(\alpha + i\beta) = 4x / (x^2 + y^2)$.

e. If $\tan(\alpha + i\beta) = x + iy$ then prove that

i. $x^2 + y^2 + 2x\cot 2\alpha = 1$
ii. $x^2 + y^2 - 2x\coth 2\beta + 1 = 0$

f. If $x + iy = 2\cosh(\alpha + \pi/4)$ then prove that $x^2 - y^2 = 2$

g. If x is real then show that $\sinh^{-1}x = \log(x+\sqrt{x^2+1})$.

h. $\tanh^{-1}x = \frac{1}{2}\log(\frac{1+z}{1-z})$.

i. $\cosh^{-1}x = \log(x+\sqrt{x^2-1})$.

j. $\tanh(log\sqrt{5}) + \tanh(log\sqrt{7}) = 17/12$.

k. $\cosh^{-1}(\sqrt{1+x^2}) = \sinh^{-1}x$

l. $\tanh^{-1}x = \sinh^{-1}(\frac{x}{\sqrt{1-x^2}})$

16. Separate into real and imaginary parts of $\sin^{-1}(3i/4)$

Logarithm of Complex numbers

17. Find the log of following complex numbers.

a. i b. 1 + i

18 Prove the following

a. $\log(1 + \cos 2x + i \sin 2x) = \log(2 \cos x) + ix$

b. $i \log(\frac{x-i}{x+i}) = \pi - 2\tan^{-1}x$

c. $\sin(\log(i^{-i})) = 1$

d. $\log(e^{i\alpha} + e^{i\beta}) = \log\{2\cos(\frac{\alpha-\beta}{2})\} + i\{\frac{\alpha+\beta}{2}\}$

e. If $i^{\alpha+i\beta} = \alpha+i\beta$ then prove that $\alpha^2 + \beta^2 = e^{-(4k+1)\pi\beta}$

f. if $i^{\log(1+i)} = A+ iB$ then prove that $A = e^{\frac{-\pi^2}{8}}\cos(\frac{\pi}{4}log2)$

g. If $i^{i} = \cos x + isin\, x\ then\ prove\ x = (4n+1)\frac{\pi}{2}e^{-(4m+1)\frac{\pi}{2}}$

h. $log\frac{\cos(x-iy)}{\cos(x+iy)} = 2itan^{-1}(\tan x \tanh y)$

i. $log\frac{\sin(x+iy)}{\sin(x-iy)} = 2itan^{-1}(\cot x \tanh y)$

j. if $\log(\log(x + iy)) = p + iq$ then $y = x\tan(\tan q.\log\sqrt{x^2+y^2})$

k. Real part of $1 + i\sqrt{3}^{1+i\sqrt{3}}$ is $2.e^{-\pi/\sqrt{3}}\cos(\frac{\pi}{3} + i\sqrt{3}\log 2)$

l. For real values of a and b $e^{-2aicot^{-1}b}.\left[\frac{bi-1}{bi+1}\right]^{-a} = 1.$

m. If $x^{x^{x^{x^{\ldots\infty}}}} = a(\cos\alpha + is\ in\ \alpha)\ then\ x = r(\cos\theta + isin\ \theta)$,
where r = {[(2nπ+α)sin α + cos α . log a]/a}.

Matrices

1. Prove that every square matrix can be uniquely expressed as sum of symmetric and skew symmetric matrix.

2. Prove that every square matrix can be uniquely expressed as sum of Hermitian and skew Hermitian matrices.

3. Prove that every square matrix can be uniquely expressed as sum p + iq, where p and q are both Hermitian matrices.

4. Express the following matrices as sum of symmetric and skew symmetric matrices.
$\begin{bmatrix}1 & 2\\ 3 & 4\end{bmatrix}$

5. Express the following matrix as sum of Hermitian and skew Hermitian matrices.
$\begin{bmatrix}1+2i & 2 & 3-i\\ 2+3i & 2i & 1-2i\\ 1+i & 0 & 3+2i\end{bmatrix}$

6. Prove that the following matrix is orthogonal.
a. $\frac{1}{3}\begin{bmatrix}1 & 2 & 2\\ 2 & 1 & -2\\ 2 & -2 & 1\end{bmatrix}$ b. $\begin{bmatrix}\cos x & 0 & \sin x\\ 0 & 1 & 0\\ -\sin x & 0 & \cos x\end{bmatrix}$ c. $\frac{1}{3}\begin{bmatrix}2 & 2 & 1\\ -2 & 1 & 2\\ -1 & 2 & -2\end{bmatrix}$

7. If A= $\frac{1}{3}\begin{bmatrix}1 & 2 & a\\ 2 & 1 & b\\ 2 & -2 & c\end{bmatrix}$ is orthogonal, then find a, b and c.

8. If A= $\frac{1}{3}\begin{bmatrix}a & b & c\\ -2 & 1 & 2\\ 1 & -2 & 2\end{bmatrix}$ is orthogonal, then find a, b and c.

9. Show that the following matrix is unitary matrix.
a. $\frac{1}{3}\begin{bmatrix}2+i & 2i\\ 2i & 2-i\end{bmatrix}$ b. $\frac{1}{2}\begin{bmatrix}1+i & -1+i\\ 1+i & 1-i\end{bmatrix}$ c. $\frac{1}{2}\begin{bmatrix}\sqrt{2} & -i\sqrt{2} & 0\\ i\sqrt{2} & -\sqrt{2} & 0\\ 0 & 0 & 2\end{bmatrix}$

10. Find the inverse of the following matrices using Adjoint method.

a. $A = \begin{bmatrix} 2 & 1 & 3 \\ 3 & 1 & 2 \\ 1 & 2 & 3 \end{bmatrix}$; b. $B = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{bmatrix}$; c. $C = \begin{bmatrix} \cos x & -\sin x & 0 \\ \sin x & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

11. Reduce the following matrices to normal form and find their rank.

a. $\begin{bmatrix} 1 & 1 & 0 \\ 2 & 2 & 0 \\ 0 & 1 & 1 \end{bmatrix}$; b. $\begin{bmatrix} 1 & 1 & -1 & 1 \\ 1 & -1 & 2 & 1 \\ 3 & 1 & 0 & 3 \end{bmatrix}$; c. $\begin{bmatrix} 1 & 0 & 2 & -2 \\ 2 & -1 & 0 & -1 \\ 1 & 0 & 2 & -1 \\ 4 & -1 & 3 & -1 \end{bmatrix}$

12. For the following matrix A, Find the value of K, for which the rank of A is 3. Also find the rank of A for the remaining values of K.

$$\begin{bmatrix} 1 & 3k & 3k+4 \\ 1 & k+4 & 4k+2 \\ 1 & 2k+2 & 3k+4 \end{bmatrix}$$

13. Reduce the following matrices to normal form and find their rank.

b. $\begin{bmatrix} 1 & 1 & 0 \\ 2 & 2 & 0 \\ 0 & 1 & 1 \end{bmatrix}$; b. $\begin{bmatrix} 1 & 1 & -1 & 1 \\ 1 & -1 & 2 & 1 \\ 3 & 1 & 0 & 3 \end{bmatrix}$; c. $\begin{bmatrix} 1 & 0 & 2 & -2 \\ 2 & -1 & 0 & -1 \\ 1 & 0 & 2 & -1 \\ 4 & -1 & 3 & -1 \end{bmatrix}$

14. Reduce the following matrices to normal PAQ form and find their rank.

a. $\begin{bmatrix} 1 & 1 & 0 \\ 2 & 2 & 0 \\ 0 & 1 & 1 \end{bmatrix}$; b. $\begin{bmatrix} 1 & 1 & -1 & 1 \\ 1 & -1 & 2 & 1 \\ 3 & 1 & 0 & 3 \end{bmatrix}$; c. $\begin{bmatrix} 1 & 0 & 2 & -2 \\ 2 & -1 & 0 & -1 \\ 1 & 0 & 2 & -1 \\ 4 & -1 & 3 & -1 \end{bmatrix}$

15. Solve the following system of equations using Gauss Jordan Method.

A. $3x + y - z = 4, \quad -2x + 3y - 4z = -1, \quad x - y + 2z = 2$

B. $3x + 2y - 2z = 6, \quad x - 2y + 3z = 6, \quad 2x + 3y + 4z = 15$

C. $x + 2y + z = 8, \quad 2x + 3y + 4z = 20, \quad 4x + 3y + 2z = 16$

D. $x + y + 2z = 4, \quad 2x + 4y + 7z = 13, \quad 3x + 5y + z = 9$

16. Solve the following system of equations using Crouts Method.

A. $3x + y - z = 4, \quad -2x + 3y - 4z = -1, \quad x - y + 2z = 2$

B. $3x + 2y - 2z = 6, \quad x - 2y + 3z = 6, \quad 2x + 3y + 4z = 15$

C. $x + 2y + z = 8, \quad 2x + 3y + 4z = 20, \quad 4x + 3y + 2z = 16$

D. $x + y + 2z = 4, \quad 2x + 4y + 7z = 13, \quad 3x + 5y + z = 9$

17. Solve the following system of equations using Gauss Seidel Method.

A. $15x + y + z = 17, \quad 2x + 15y + z = 18, \quad x + 2y + 15z = 18$

B. $10x + 2y + z = 9, \quad 2x + 20y - 2z = -44, \quad -2x + 3y + 10z = 22$

C. $82x - 3y + z = 75, \quad x + 75y - 2z = 153, \quad 3x - 2y + 85z = -86$
D. $2x + y - z = 40, \quad 2x + 18y + z = 21, \quad x + 2y + 25z = -21$

Matrices- System of Linear Equations

1. Determine the consistency of the following system of equations and if possible find their solution. (Non-Homogeneous and Homogenous cases)

A. $3x + y + 2z = 11, \quad 2x + 3y + z = 11, \quad x + 2y + 3z = 14$
B. $2x + y + z = 4, \; x - y + 3z = 3, \; 4x - y - z = 2$
C. $x - 3y - 8z = -10, \; 3x + y - 4z = 0, \; 2x + 5y + 6z = 13$
D. $4x - 2y + 6z = 8, \quad x + y - 3z = -1, \; 15x - 3y + 9z = 21$
E. $x - y + z = 0, \quad x + 2y + z = 0, \; 2x + y + 3z = 0$
F. $7x_1 + x_2 - 2x_3 = 0, \quad x_1 + 5x_2 - 4x_3 = 0,$
$3x_1 - 2x_2 + x_3 = 0, \; 2x_1 - 7x_2 + 5x_3 = 0$

2. Investigate for what value of λ and μ the equations below has (i) No solution, (ii) Unique solution, (iii) Infinitely many solutions.

$$2x + 3y + 5z = 9, \; 7x + 3y - 2z = 8, \quad 2x + 3y + \lambda z = \mu$$

3. Investigate for what value of λ, the equations below has a solution and solve them in each case.

$$x + 2y + z = 3, \; x + y + z = \lambda, \quad 3x + y + 3z = \lambda^2$$

4. Show that the system of equations below can possess a non-trivial solution only if $\lambda = 1$, or $\lambda = -3$. Obtain the general solution in each case.

$$2x - 2y + z = \lambda x, \; 2x - 3y + 2z = \lambda y, \quad -x + 2y = \lambda z$$

5. For what value of λ the system of equations below can possess a non-trivial solution. Obtain the general solution for real value of λ.

$$3x + y - \lambda z = 0, \; 4x - 2y - 3z = 0, \quad 2\lambda x + 4y + \lambda z = 0$$

Partial derivatives

1 If $u = \log(x^2 + y^2)$ then prove that $\frac{\partial^2 u}{\partial x \partial y} = \frac{\partial^2 u}{\partial y \partial x}$. 5 or 6

2 If $u = (1 - 2xy + y^2)^{-1/2}$ prove that $x\frac{\partial u}{\partial x} - y\frac{\partial u}{\partial x} = y^2u^3$. 5 or 6

3 If $u = \log(\tan x + \tan y)$, prove that $\sin 2x\frac{\partial u}{\partial x} + \sin 2y\frac{\partial u}{\partial y} = 2$. 5 or 6

4 If $u = lx + my$, $v = mx - ly$, prove that 8 or 10

(i) $\left(\frac{\partial u}{\partial x}\right)_y \times \left(\frac{\partial x}{\partial u}\right)_v = \frac{l^2}{l^2+m^2}$

(ii) $\left(\frac{\partial y}{\partial v}\right)_x \times \left(\frac{\partial v}{\partial y}\right)_u = \frac{l^2+m^2}{l^2}$

5 If $x^2 = au + bv$, $y^2 = au - bv$, prove that 8 or 10

$\left(\frac{\partial u}{\partial x}\right)_y \times \left(\frac{\partial x}{\partial u}\right)_v = \frac{1}{2} = \left(\frac{\partial v}{\partial y}\right)_x \times \left(\frac{\partial y}{\partial v}\right)_u$

6 If $z = \sin^{-1}(x - y)$, $x = 3t, y = 4t^3$ prove that $\frac{dz}{dt} = \frac{3}{\sqrt{1-t^2}}$. 5 or 6

7 If $z = \tan^{-1}\left(\frac{x}{y}\right)$, $x = 2t, y = 1 - t^2$ prove that $\frac{dz}{dt} = \frac{2}{1+t^2}$. 5 or 6

8 If $z = e^{xy^2}$, $x = t\cos t$, $y = t\sin t$, find $\frac{dz}{dt}$ at $t = \frac{\pi}{2}$. 5 or 6

9 If $u = f(r^2)$ and $r^2 = x^2 + y^2 + z^2$, prove that 5 or 6

$u_{xx} + u_{yy} + u_{zz} = 4\mathrm{r}^2\mathrm{f}''(\mathrm{r}^2) + 6\mathrm{f}'(\mathrm{r}^2)$

10 If $z = x\log(x + r) - r$, where $r^2 = x^2 + y^2$, prove that 8 or 10

$\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = \frac{1}{x+r}$, or $\frac{\partial^3 z}{\partial x^3} = -\frac{x}{r^3}$

11 If $u = \frac{1}{r}$ and $r^2 = x^2 + y^2 + z^2$, prove that 8 or 10

(i) $xu_x + yu_y + zu_z + u = 0$.

(ii) $u_{xx} + u_{yy} + u_{zz} = 0$.

12 If $u = e^{xyz} f\left(\frac{xy}{z}\right)$, prove that 8 or 10

(i) $xu_x + zu_z = 2xyzu$

(ii) $yu_y + zu_z = 2xyzu$

13 If $u = x^2 - y^2$, $v = 2xy$ and $z = f(u, v)$, prove that 5 or 6

$\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 = 4\sqrt{u^2 + v^2}\left[\left(\frac{\partial z}{\partial u}\right)^2 + \left(\frac{\partial z}{\partial v}\right)^2\right]$

14 Verify Euler's theorem for $= x^2 \tan^{-1}\frac{y}{x} - y^2 \tan^{-1}\frac{x}{y}$. 8 or 10

15 Using Euler's theorem find $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y}$, if $u = \frac{\sqrt{x}+\sqrt{y}}{x+y}$. 5 or 6

16 If $u = \frac{x^2y^2}{x^2+y^2} + \cos\left(\frac{xy}{x^2+y^2}\right)$, prove that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = 2\left(\frac{x^2y^2}{x^2+y^2}\right)$. 8 or 10

17 If $u = x^2 \tan^{-1}\left(\frac{y}{x}\right) + y^2 \sin^{-1}\left(\frac{x}{y}\right)$, prove that 5 or 6

$x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x \partial y} + y^2\frac{\partial^2 u}{\partial y^2} = 2u$.

18 If $u = x^3 \sin^{-1}\left(\frac{y}{x}\right) + x^4\, tan^{-1}\left(\frac{y}{x}\right)$, find the value of 8 or 10

$x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x \partial y} + y^2\frac{\partial^2 u}{\partial y^2} + x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y}$ at (1,1) .

19 If $u = \sin^{-1}\left[\frac{x^{\frac{1}{4}}+y^{\frac{1}{4}}}{x^{\frac{1}{5}}+y^{\frac{1}{5}}}\right]$, prove that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = \frac{1}{20}\tan u$. 5 or 6

20 If $u = cosec^{-1}\left(\sqrt{\frac{x^{\frac{1}{2}}+y^{\frac{1}{2}}}{x^{\frac{1}{3}}+y^{\frac{1}{3}}}}\right)$, Prove that 8 or 10

$x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x \partial y} + y^2\frac{\partial^2 u}{\partial y^2} = \frac{\tan u}{144}(13 + \tan^2 u)$

21 If $u = \log(x^3 + y^3 - x^2y - xy^2)$, prove that 5 or 6

$x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x \partial y} + y^2\frac{\partial^2 u}{\partial y^2} = -3$

22 If $u = \frac{x^2y^2}{x^2+y^2} + \cos^{-1}\left(\frac{x+y}{\sqrt{x}+\sqrt{y}}\right)$, find the value of $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y}$ 5 or 6

Applications of Partial differentiation

1	Find the stationary points and extreme values of $x^3 + 3xy^2 - 15x^2 - 15y^2 + 72x$.	5 or 6
2	Find the stationary points and extreme values of $xy(3 - x - y)$.	5 or 6
3	Find the stationary points and extreme values of $\sin x \sin y \sin(x + y)$.	5 or 6
4	Find the stationary points and extreme values of $x^3 + y^3 - 63(x + y) + 12xy$.	5 or 6
5	Find the stationary points and extreme values of $x^3y^2(12 - 3x - 4y)$.	5 or 6
6	Find the value of $x^2 + y^2$ subjected to the condition $x + y = 2$ by Lagrange's method.	5 or 6
7	Find the points on the ellipse$x^2 + 2y^2 = 1$, at which $f(x, y) = xy$ takes extreme values.	5 or 6
8	Find the points on the curve $xy^2 = 54$ which are nearest to the origin.	5 or 6
9	Find the minimum distance of the origin from the plane $3x + 2y + z - 12 = 0$.	8 or 10
10	Find the maximum and minimum distances of the point $(3,4,12)$ from the sphere $x^2 + y^2 + z^2 = 1$.	8 or 10
11	Show that the rectangular solid of maximum volume that can be inscribed in a sphere is a cube.	8 or 10
12	The sum of three positive numbers is 1. Determine the maximum value of their product.	5 or 6
13	If 120 is divided into three numbers such that the sum of their product taken two at a time is maximum, find the numbers.	5 or 6

Successive Differentiation

14	Find the nth derivative of $\frac{x^2}{(x-1)^2(x+2)}$.	5 or 6
15	If $y = x \log\left(\frac{x-1}{x+1}\right)$, prove that $y_n = (-1)^{n-2}(n-2)!\left[\frac{x-n}{(x-1)^n} - \frac{x+n}{(x+1)^n}\right]$.	8 or 10
16	If $y = 2^x \sin^2 x \cos^3 x$, find y_n.	8 or 10
17	If $y = \tan^{-1}\left(\frac{1+x}{1-x}\right)$, prove that $y_n = (-1)^{n-1}(n-1)! \sin^n\theta \sin n\theta$.	5 or 6
18	If $y = e^{msin^{-1}x}$ or $x = \sin\left(\frac{1}{m} log\, y\right)$, prove that $(1 - x^2)y_{n+2} - (2n + 1)xy_{n+1} - (n^2 + m^2)y_n = 0$.	5 or 6
19	If $y = 2x\sqrt{1 - x^2}$, prove that $(1 - x^2)y_{n+2} - (2n + 1)xy_{n+1} - (n^2 - 4)y_n = 0$.	5 or 6
20	If $y = \sin \log(x^2 + 2x + 1)$, prove that $(x + 1)^2y_{n+2} + (2n + 1)(x + 1)y_{n+1} + (n^2 + 4)y_n = 0$.	8 or 10
21	If $y = \tan^{-1}\left[\frac{a+x}{a-x}\right]$, prove that $(a^2 + x^2)y_{n+2} + 2(n + 1)xy_{n+1} + n(n + 1)y_n = 0$.	5 or 6
22	If $y = e^{\text{asin}^{-1}x}$, prove that	8 or 10

(i) $(1-x^2)y_{n+2}-(2x+1)xy_{n+1}-(x^2+a^2)y_n=0$
(ii) $y_{2m}(0)=a(a^2+2^2)(a^2+4^2)(a^2+6^2)\ldots.(a^2+(2m-2)^2)$.

1. Find the nth Derivative of following functions.

   a. $\frac{x}{x^2+a^2}$. b. $\frac{x^2}{(x+2)(2x+3)}$.

   c. $\frac{x^4}{(x-1)(x-2)}$. d. $\frac{x^2}{(x-1)(2x+3)}$.

   e. $\sin 2x\cos 6x$. f. $\sin x\cos 3x$.

   g. $\sin 2x sin\ 3x cos\ 4x$ h. $\sin 2x\sin 3x\sin 4x$

2. a) If $y=x^n$ log x, then prove that $y_{n+1}=\frac{n!}{x}$ using Leibnitz's Rule.

   b) If $=x^2e^x$, then prove that at x = 0, $y_n=2^{(n-2)}n(n-1)$. using Leibnitz's Rule.

   c) If $y=(2x+3)^2e^x$ then find y_n.